

Communication

The devices that make up a computer all communicate in binary by moving bytes of data. Everything they “say” to each other is made up of a string of 0s and 1s.

Bytes of data move back and forth constantly.

Processor

The central processing unit (CPU) is the “brain” of a computer, sending and receiving bytes of data. This carries out calculations and instructions.

Hard drive

The hard drive is the main storage area for billions of 0s and 1s. This stores the bytes even while the computer is off.

All data stored is a series of electrical pulses that are either on, “1” or off, “0.”

RAM

RAM means Random Access Memory. Bytes of data are stored here while the computer is on. The data is lost if the user has not saved it.

STORING DATA

How many bytes?

Computer storage is described using bytes. These units show how many 0s and 1s the computer has space for.

- 1 1 **kilobyte** is 1,024 Bytes
- 2 1 **megabyte** is 1,024 Kilobytes
- 3 1 **gigabyte** is 1,024 Megabytes
- 4 1 **terabyte** is 1,024 Gigabytes
- 5 1 **petabyte** is 1,024 Terabytes
- 6 1 **exabyte** is 1,024 Petabytes

ASCII

To turn binary numbers into characters people recognize, computers use ASCII (“askey”), the American Standard Code for Information Interchange. Each set of eight binary numbers represents a number, letter, or symbol in ASCII code.

Binary	ASCII
01000001	A
01000010	B
01000011	C
00110001	1
00110010	2
00110011	3
00101110	.
00100001	!
00111111	?
01000000	@